

RAM AND STORAGE THEORY.

- 1) Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

DDR3 vs DDR4 vs DDR5 RAM Comparison

Feature	DDR3	DDR4	DDR5
Release Period	2007	2014	2020
Typical Speed	800–2133 MT/s	1600–3200 MT/s	4800–8400+ MT/s
Voltage	1.5V (1.35V for DDR3L)	1.2V	1.1V
Maximum Module Capacity	Usually up to 16 GB	Usually up to 64 GB	128 GB+ possible
Bandwidth	Lowest	Higher than DDR3	Highest
Power Efficiency	Lowest	Better	Best
Physical Compatibility	Unique slot/key notch	Different notch from DDR3	Different notch from DDR4
Typical Use Cases	Older desktops/laptops, legacy systems	Mainstream PCs (2015–2023)	Modern gaming PCs, AI workloads, content creation, newest laptops/desktops

- 2) Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.

- Remove ram module from motherboards :



- instal ram module from motherboard



- 3) List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Feature	HDD	SATA SSD	NVMe SSD
Storage Technology	Mechanical disks	Flash memory	Flash memory over PCIe
Typical Speed	80–200 MB/s	500–600 MB/s	1500–14000+ MB/s
Moving Parts	Yes	No	No
Noise	Audible	Silent	Silent
Cost per GB	Lowest	Moderate	Highest
Best For	Large cheap storage	Everyday computing	High-performance workloads

- 4) Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after

reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.

